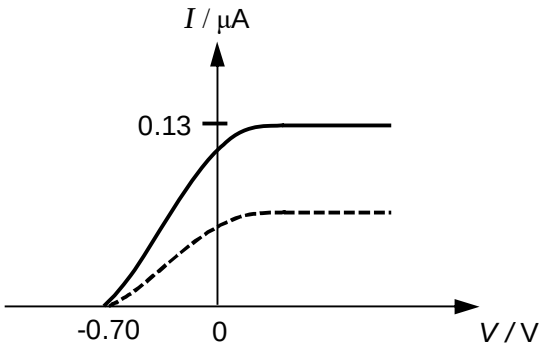


6(a)	It is the phenomenon whereby electrons from a metal are emitted when electromagnetic radiation of sufficiently high frequency is incident on the metal.	[1]
(b)(i)	From the graph, stopping potential is 0.70 V (not -0.70V). Hence, maximum kinetic energy of the photoelectrons = $eV_s = 1.12 \times 10^{-19} \text{ J}$	[1]
(ii)	$hf = \phi + E_{k \max} \rightarrow f = \frac{(2.06)(1.60 \times 10^{-19}) + 1.12 \times 10^{-19}}{6.63 \times 10^{-34}}$ $= 6.66 \times 10^{14} \text{ Hz}$	[1] Sub [1] ans
(iii)	For the photoelectrons, the saturated current $I = \frac{Q}{t} = \frac{Ne}{t}$ → No. of electrons emitted per second: $\frac{N}{t} = \frac{I}{e} = \frac{0.13 \times 10^{-6}}{1.60 \times 10^{-19}} = 8.13 \times 10^{11} \text{ s}^{-1}$ Number of photons incident per second: $\frac{N_{\text{photon}}}{t} = (8.13 \times 10^{11})(8000) = 6.50 \times 10^{15} \text{ s}^{-1}$ Intensity = $\frac{E}{tA} = \frac{N_{\text{photon}}(hf)}{tA} = \left(\frac{N_{\text{photon}}}{t} \right) \left(\frac{hf}{A} \right)$ $= (6.50 \times 10^{15}) \frac{(6.63 \times 10^{-34})(6.66 \times 10^{14})}{1.0 \times 10^{-4}} = 28.7 \text{ W m}^{-2}$	[1] No of electrons s^{-1} [1] No of photons s^{-1} [1] ans
MC	Ecf within part	

6(b)(iv)	 Fig. 1.1	[1]
(v)	The negative p.d. creates a repulsion to the emitted photoelectrons causing less electrons to reach the collector. (effect) It denotes the fact that the electrons are emitted with a range of KE. (cause)	[1] [1]
7(a)(i)	$A \equiv 94 \quad Z \equiv 35$	[1]